Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |

AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

AI agent engineer who **built AgentProbe end-to-end** - production LangGraph agent platform with prompt engineering, dynamic model selection, custom tool calls, and a 5-metric evaluation engine that enables faster iteration loops. Designed agent environments, optimized prompts against ground-truth evaluation, and built the measurement systems that catch quality regressions before users see them. 8+ years production Python; domain-specific AI (healthcare NLP via AI on FHIR); applied ML on 1.6M financial records at Entrata. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

LLM API & Agent Frameworks (LangChain/LangGraph) | Prompt Engineering & Optimization | Model Selection & Multi-Provider Routing | Agent Environment Design | Evaluation Design & LLM-as-Judge | Context Window Management | Tool Call Architecture | RAG + Vector Search | Production AI Systems (FastAPI) | Domain-Specific AI (Healthcare / Legal Patterns) | Python (8+ years) | CI/CD Eval Integration

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Applied **Python, PyTorch, and OpenAI API** to automate domain-specific data transformation across **1.6M financial records** - production AI applied to regulated data workflows, requiring careful prompt design and output validation.
- Built and delivered **20+ full-stack modules** (Python, JavaScript, PostgreSQL) across **120+ Agile sprints** for **100K+ monthly active users**; drove cross-functional alignment across Product, Engineering, and QA.
- Maintained **99.5% uptime** for 6+ months via TDD/BDD; improved production query performance by **30%**; built **5+ debugging tools** reducing resolution time by **40%**.
- Migrated client data to AWS S3 with encrypted pipelines - **zero security incidents** across 100K+ users; enforced secure coding standards across all major releases.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students in Python and databases; delivered targeted feedback that accelerated concept mastery - the iterative, quality-focused loop that agent eval systems replicate at scale.

PROJECTS

AgentProbe: AI Agent Platform with Evaluation Engine OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Production LangGraph agent platform built to make AI systems measurably better through systematic evaluation - the exact discipline Harvey's Agents team applies to legal workflows. **Agent environment design**: 3-agent system (Orchestrator, RAG Tracer, Auto Fix) with defined action spaces, tool sets, and decision boundaries per agent. **Model selection**: dynamic routing across Anthropic, OpenAI, Groq, Ollama - selecting provider based on task, latency, and capability. **Prompt engineering**: calibrated judge prompts for 5-metric evaluation (factual accuracy, hallucination, retrieval, completeness, compliance); iterative optimization against reference outputs. **Evaluation design**: LLM-as-judge with regression tracking and CI/CD gates - the feedback loop that makes agents better with every iteration. **Context management**: multi-step agent runs with state preservation across tool calls. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: Domain-Specific NLP for Medical Records

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

NLP pipeline mapping natural language queries to domain-specific FHIR APIs via LLMs, BERT, and embeddings - HIPAA-compliant RBAC. **~60% retrieval time reduction.** Demonstrates building AI agents for professional workflows in regulated domains - the same pattern Harvey applies to legal. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection (RF vs LR) → rigorous evaluation (MDI + Permutation Importance) → deployment + stakeholder dashboard. **85%+ accuracy.** Systematic model selection and evaluation methodology.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

LLM & Agents: LangChain, LangGraph, LLM APIs (Anthropic, OpenAI, Groq, Ollama), Prompt Engineering, Model Selection, Agent Environment Design, Tool Call Architecture, Context Window Management, RAG Pipelines, ChromaDB, LLM-as-Judge Evaluation, FastAPI

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Cloud & DevOps: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), PostgreSQL, MySQL, React

Process: TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum